

Exercise #9

13. March 2023

Problem 1.

In this problem, we consider the numerical solution of the heat equation

$$u_t = u_{xx} \quad \text{for } 0 < x < 1 \text{ and } t > 0$$

with the initial condition

$$u(x, 0) = f(x) \quad \text{for } 0 \leq x \leq 1$$

for some given $f: [0, 1] \rightarrow \mathbb{R}$, and the mixed boundary conditions

$$\partial_x u(0, t) = a(u(0, t) - g(t)) \quad \text{and} \quad \partial_x u(1, t) = p(t) \quad \text{for } t > 0,$$

where $a > 0$ is some fixed parameter, $g: \mathbb{R}_{>0} \rightarrow \mathbb{R}$ and $p: \mathbb{R}_{>0} \rightarrow \mathbb{R}$ are given functions.

This models for instance the temperature in a thin, insulated rod, with heat transfer through both endpoints. At $x = 0$, we can understand $g(t)$ as describing the outside temperature and a as the heat transfer coefficient. At $x = 1$, $p(t)$ could model a controlled heating or cooling of the rod endpoint.

- a) Set up an *explicit* finite difference scheme for the solution of this PDE. The final scheme should be written in the form

$$\begin{aligned} U_i^0 &= \dots & \text{for } i = 0, \dots, M, \\ U_0^{n+1} &= \dots & \text{for } n = 0, 1, \dots, \\ U_i^{n+1} &= \dots, & \text{for } i = 1, \dots, M-1, \text{ and } n = 0, 1, \dots, \\ U_M^{n+1} &= \dots, & \text{for } n = 0, 1, \dots, \end{aligned}$$

where the ... consist of given and/or previously computed values.

- b) Set up an *implicit* finite difference scheme for the solution of this PDE. The final scheme should be written in matrix-vector form as

$$\begin{aligned} U_i^0 &= \dots & \text{for } i = 0, \dots, M, \\ C\vec{U}^{n+1} &= D\vec{U}^n + \dots & \text{for } n = 0, 1, \dots, \end{aligned}$$

where $C, D \in \mathbb{R}^{(M+1) \times (M+1)}$ are matrices and ... consist of given values.

- c) Consider in particular the case where $a = \frac{\pi}{2}$, $g(t) = \frac{3}{2}e^{-\frac{\pi^2}{4}t}$ and $p(t) = -\frac{\pi}{2}e^{-\frac{\pi^2}{4}t}$ for all $t > 0$, and

$$f(x) = \left(\cos\left(\frac{\pi}{2}x\right) - \frac{1}{2} \sin\left(\frac{\pi}{2}x\right) \right).$$

Verify that the function

$$u(x, t) = \left(\cos\left(\frac{\pi}{2}x\right) - \frac{1}{2} \sin\left(\frac{\pi}{2}x\right) \right) e^{-\frac{\pi^2}{4}t}$$

is a solution of the PDE with these boundary and initial conditions.

- d) (J) Implement the schemes from parts [a\)](#) and [b\)](#), and test them for the initial and boundary conditions of part [c\)](#). Plot your numerical approximations for $t = 0, 0.1, 1.0$, and 5.0 , and compare them with the exact solution.

Consider specifically your explicit scheme with step size $h = 0.1$. For which time step length k (approximately) does the solution become unstable?

You may use the jupyter templates that are provided together with the exercise sheet.

Note: The main task in parts [a\)](#) and [b\)](#) of this exercise is to find a correct way of incorporating the mixed boundary conditions in the numerical methods. Your answer should include a detailed explanation of how you treat these boundary conditions, both in the explicit and the implicit case. The other steps in the discretisation, which are really the same as in the lecture / lecture notes can be less detailed.

Solution.

- a) Let M be the number of discretisation points between $[0, 1]$, and let $h = \frac{1}{M}$ be the step length in x -direction. The gridpoints are defined by $x_i = ih$, $i = 0, \dots, M$. Let k be the step length in t -direction, and define the gridpoints $t_n = nk$, $n = 0, 1, \dots$

We use a forward difference to approximate u_t and a second order difference for u_{xx} . Let the numerical approximation be $U_i^n = u(x_i, t_n)$, then we obtain as in the lecture note

$$\frac{U_i^{n+1} - U_i^n}{k} = \frac{U_{i+1}^n - 2U_i^n + U_{i-1}^n}{h^2}.$$

Rearranging the terms, we get

$$U_i^{n+1} = U_i^n + \frac{k}{h^2} (U_{i+1}^n - 2U_i^n + U_{i-1}^n). \quad (1)$$

We now have a scheme, but we need to handle the boundary conditions. On the right hand side, we have a Neumann boundary condition, (1) is here given by

$$U_M^{n+1} = U_M^n + \frac{k}{h^2} (U_{M+1}^n - 2U_M^n + U_{M-1}^n). \quad (2)$$

We need to find an approximation for U_{M+1}^n . Using a central difference to approximate $\partial_x u(1, t)$ gives

$$\begin{aligned} \frac{U_{M+1}^n - U_{M-1}^n}{2h} &= p(t_n), \\ U_{M+1}^n &= U_{M-1}^n + 2hp(t_n). \end{aligned}$$

By inserting this into (2), we end up with

$$U_M^{n+1} = U_M^n + \frac{k}{h^2} (-2U_M^n + 2U_{M-1}^n + 2hp(t_n)).$$

On the left hand side, we have a Robin boundary condition. Here, (1) is of the form,

$$U_0^{n+1} = U_0^n + \frac{k}{h^2} (U_1^n - 2U_0^n + U_{-1}^n). \quad (3)$$

As for the Neumann condition, we use a central difference to approximate $\partial_x u(0, t)$,

$$\frac{U_1^n - U_{-1}^n}{2h} = a (U_0^n - g(t_n)).$$

Solving for U_{-1}^n gives

$$U_{-1}^n = U_1^n - 2ah (U_0^n - g(t_n)).$$

Inserting this into (3) gives

$$U_0^{n+1} = U_0^n + \frac{k}{h^2} (2U_1^n - 2U_0^n - 2ah (U_0^n - g(t_n))) .$$

Finally, we need the initial conditions, which are used to initialize, $U_i^0 = f(x_i)$. Summing up, the total explicit scheme is,

$$\begin{aligned} U_i^0 &= f(x_i) & \text{for } i = 0, \dots, M, \\ U_0^{n+1} &= U_0^n + \frac{k}{h^2} (2U_1^n - 2U_0^n - 2ah (U_0^n - g(t_n))) & \text{for } n = 0, 1, \dots, \\ U_i^{n+1} &= U_i^n + \frac{k}{h^2} (U_{i+1}^n - 2U_i^n + U_{i-1}^n) & \text{for } i = 1, \dots, M-1, \text{ and } n = 0, 1, \dots, \\ U_M^{n+1} &= U_M^n + \frac{k}{h^2} (-2U_M^n + 2U_{M-1}^n + 2hp(t_n)), & \text{for } n = 0, 1, \dots \end{aligned}$$

b) For the implicit scheme, we use a backward scheme to approximate u_t ,

$$\frac{U_i^{n+1} - U_i^n}{k} = \frac{U_{i+1}^{n+1} - 2U_i^{n+1} + U_{i-1}^{n+1}}{h^2} .$$

Gathering all terms involving U_*^{n+1} on the left side gives

$$U_i^{n+1} - \frac{k}{h^2} (U_{i+1}^{n+1} - 2U_i^{n+1} + U_{i-1}^{n+1}) = U_i^n . \quad (4)$$

On the right hand side, (4) is given by,

$$U_M^{n+1} - \frac{k}{h^2} (U_{M+1}^{n+1} - 2U_M^{n+1} + U_{M-1}^{n+1}) = U_M^n .$$

By again using the central difference, we get

$$U_M^{n+1} - \frac{k}{h^2} (-2U_M^{n+1} + 2U_{M-1}^{n+1} + 2hp(t_{n+1})) = U_M^n .$$

Where we move the boundary condition to the right-hand side to obtain

$$U_M^{n+1} - \frac{k}{h^2} (-2U_M^{n+1} + 2U_{M-1}^{n+1}) = U_M^n + \frac{2k}{h} p(t_{n+1}) .$$

On the left hand side, (4) is given by,

$$U_0^{n+1} - \frac{k}{h^2} (U_1^{n+1} - 2U_0^{n+1} + U_{-1}^{n+1}) = U_0^n ,$$

and we also here use the central difference,

$$\frac{U_1^{n+1} - U_{-1}^{n+1}}{2h} = a (U_0^{n+1} - g(t_{n+1})).$$

Solving for U_{-1}^{n+1} gives

$$U_{-1}^{n+1} = U_1^{n+1} - 2ah (U_0^{n+1} - g(t_{n+1})),$$

and inserting into (4) gives

$$U_0^{n+1} - \frac{k}{h^2} (U_1^{n+1} - 2U_0^{n+1} + U_{-1}^{n+1} - 2ahU_0^{n+1}) = U_0^n - \frac{2ak}{h} g(t_{n+1}).$$

The initial conditions are handled the same way as for the explicit scheme. We then have the final scheme

$$\begin{aligned} U_i^0 &= f(x_i) & \text{for } i = 0, \dots, M, \\ CU^{n+1} &= U^n + b^{n+1} & \text{for } n = 0, 1, \dots, \end{aligned}$$

where $C = I + rA$, with I as the identity matrix and

$$A = \begin{pmatrix} 2(1+ah) & -1 & 0 & \dots & \dots & \dots & 0 \\ -1 & 2 & -1 & 0 & \dots & \dots & 0 \\ 0 & -1 & 2 & -1 & 0 & \dots & 0 \\ \vdots & \ddots & \ddots & \ddots & \ddots & \ddots & \vdots \\ \vdots & & \ddots & \ddots & \ddots & \ddots & \vdots \\ 0 & \dots & \dots & 0 & -1 & 2 & -1 & 0 \\ 0 & \dots & & \dots & 0 & -1 & 2 & -1 \\ 0 & \dots & & & \dots & 0 & -2 & 2 \end{pmatrix}, \quad (5)$$

with $r = \frac{k}{h^2}$, and

$$b^{n+1} = \begin{pmatrix} \frac{2ak}{h} g(t_{n+1}) \\ 0 \\ \vdots \\ 0 \\ \frac{2k}{h} p(t_{n+1}) \end{pmatrix}. \quad (6)$$

c) We first check the initial condition,

$$u(x, 0) = \left(\cos\left(\frac{\pi}{2}x\right) - \frac{1}{2} \sin\left(\frac{\pi}{2}x\right) \right) = f(x).$$

Then, we calculate the derivatives

$$u_x(x, t) = \frac{\pi}{2} \left(-\sin\left(\frac{\pi}{2}x\right) - \frac{1}{2} \cos\left(\frac{\pi}{2}x\right) \right) e^{-\frac{\pi^2}{4}t}$$

$$u_{xx}(x, t) = -\frac{\pi^2}{4} \left(\cos\left(\frac{\pi}{2}x\right) - \frac{1}{2} \sin\left(\frac{\pi}{2}x\right) \right) e^{-\frac{\pi^2}{4}t}$$

$$u_t(x, t) = -\frac{\pi^2}{4} \left(\cos\left(\frac{\pi}{2}x\right) - \frac{1}{2} \sin\left(\frac{\pi}{2}x\right) \right) e^{-\frac{\pi^2}{4}t}$$

We see that $u_t = u_{xx}$. The last thing to check is the boundary conditions.

$$u_x(0, t) = -\frac{\pi}{4} e^{-\frac{\pi^2}{4}t}$$

$$a(u(0, t) - g(t)) = \frac{\pi}{2} \left(e^{-\frac{\pi^2}{4}t} - \frac{3}{2} e^{-\frac{\pi^2}{4}t} \right) = -\frac{\pi}{4} e^{-\frac{\pi^2}{4}t}.$$

And similarly

$$u_x(1, t) = -\frac{\pi}{4} e^{-\frac{\pi^2}{4}t} = p(t).$$

This shows that the boundary conditions are satisfied, meaning that $u(x, t)$ is a solution of the PDE.

- d) See the Jupyter notebooks `heat_numerics_explicit.ipynb`, `heat_numerics_implicit.ipynb` for implementation. By testing, we can see that for the explicit scheme with $h = 0.1$, the solution becomes unstable if we set $k = 0.005$ or higher. The CFL-condition also gives that when $k > 0.005$ we have $r > 0.5$. For the implicit scheme, we do not experience any instability.

Problem 2.

Use the Fourier transform to find the function $u(x, t)$ satisfying the PDE

$$u_t = tu_{xx} \quad \text{for } x \in \mathbb{R} \text{ and } t > 0$$

with the initial condition

$$u(x, 0) = e^{-2x^2} \quad \text{for } x \in \mathbb{R}.$$

Solution.

We do a Fourier transform (in x) of the PDE,

$$\begin{aligned} \widehat{u}_t &= -t\omega^2 \widehat{u}, \\ \widehat{u}(\omega, 0) &= \frac{1}{2}e^{-\frac{\omega^2}{8}}. \end{aligned}$$

For the initial condition, we have used the Fourier transformation (can be found in the table in Krezig)

$$\mathcal{F}(e^{-ax^2}) = \frac{1}{\sqrt{2a}}e^{-\frac{\omega^2}{4a}}. \quad (7)$$

We now have the problem in the form of an ODE. We solve by separation of variables

$$\begin{aligned} \frac{d\widehat{u}}{dt} &= -t\omega^2 \widehat{u} \\ \frac{d\widehat{u}}{\widehat{u}} &= -t\omega^2 dt \\ \ln|\widehat{u}| &= -\frac{1}{2}t^2\omega^2 + C \\ \widehat{u} &= C(\omega)e^{-\frac{1}{2}t^2\omega^2}, \end{aligned}$$

and find C from the initial condition,

$$\widehat{u}(0) = C = \frac{1}{2}e^{-\frac{\omega^2}{8}}.$$

The solution to the ODE is then

$$\widehat{u}(t, \omega) = \frac{1}{2}e^{-\frac{\omega^2}{2}(\frac{1}{4}+t^2)}.$$

We now need to do an inverse Fourier transformation, and again we use (7), with $a = \frac{1}{\frac{1}{2} + 2t^2}$, which leads to the function $u(x, t)$,

$$u(x, t) = \frac{1}{2\sqrt{1+4t^2}} e^{-\frac{x^2}{\frac{1}{2} + 2t^2}}.$$

Problem 3.

Use the Fourier transform to find the function $u(x, t)$ satisfying the *wave equation* on an infinite string

$$u_{tt} = u_{xx} \quad \text{for } -\infty < x < \infty \text{ and } t > 0$$

with the initial conditions

$$\left. \begin{aligned} u(x, 0) &= \frac{1}{x^2 + 1}, \\ \partial_t u(x, 0) &= 0 \end{aligned} \right\} \quad \text{for } x \in \mathbb{R}.$$

Solution.

We do a Fourier transform (in x) of the PDE,

$$\widehat{u}_{tt} = -\omega^2 \widehat{u}. \quad (8)$$

And use the following Fourier transform (can be found in the table in Krezig) to transform the initial condition

$$\mathcal{F}\left(\frac{1}{x^2 + 1}\right) = \sqrt{\frac{\pi}{2}} e^{-|\omega|} \quad (9)$$

Considering the transformed equation (8), the general solution is given by

$$\widehat{u}(\omega, t) = A(\omega) \cos(\omega t) + B(\omega) \sin(\omega t).$$

Since $\widehat{u}_t(\omega, 0) = 0$, we have to require that $B(\omega) \equiv 0$. For the other condition, we find that

$$\widehat{u}(\omega, 0) = A(\omega) = \sqrt{\frac{\pi}{2}} e^{-|\omega|}$$

Which gives a solution of the form

$$\widehat{u}(\omega, t) = \widehat{u}(\omega, 0) \cos(\omega t).$$

Recall now the Euler formula, that allows us to write

$$\cos(\omega t) = \frac{1}{2}(e^{i\omega t} + e^{-i\omega t}).$$

For the inverse Fourier transform, we find by using the Euler formula:

$$\begin{aligned} u(x, t) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \widehat{u}(\omega, 0) \cos(\omega t) e^{i\omega x} d\omega \\ &= \frac{1}{2\sqrt{2\pi}} \int_{-\infty}^{\infty} \widehat{u}(\omega, 0) (e^{i\omega t} + e^{-i\omega t}) e^{i\omega x} d\omega \\ &= \frac{1}{2\sqrt{2\pi}} \int_{-\infty}^{\infty} \widehat{u}(\omega, 0) (e^{i\omega(x+t)} + e^{i\omega(x-t)}) d\omega \\ &= \frac{1}{2} \left(u(x+t, 0) + u(x-t, 0) \right) \\ &= \frac{1}{2} \left(\frac{1}{(x+t)^2 + 1} + \frac{1}{(x-t)^2 + 1} \right). \end{aligned}$$

Where the fourth equality comes from the fact that the inverse transform of $\widehat{u}(\omega, 0)$ is given by

$$u(x, 0) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \widehat{u}(\omega, 0) e^{i\omega x} d\omega.$$

One might notice that this solution is readily found by applying D'Alembert's solution of the wave equation. However, the point of this task is to apply the Fourier transform and the Euler formula. If the students directly solve the problem using D'Alembert, they should get a slightly reduced score.

The next exercises are optional and should not be handed in!

Problem 4.

Use the Fourier transform to find the function $u(x, t)$ satisfying the *telegraph equation*

$$u_{tt} + 2u_t + u = u_{xx} \quad \text{for } x \in \mathbb{R} \text{ and } t > 0$$

with the initial conditions

$$\left. \begin{aligned} u(x, 0) &= \text{sinc}(x), \\ \partial_t u(x, 0) &= -\text{sinc}(x) \end{aligned} \right\} \quad \text{for } x \in \mathbb{R}.$$

Here the sinc function is defined as

$$\text{sinc}(x) = \begin{cases} \frac{\sin(x)}{x} & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$$

Solution.

We do a Fourier transform (in x) of the PDE,

$$\widehat{u}_{tt} + 2\widehat{u}_t + \widehat{u} = -\omega^2 \widehat{u},$$

and write the ODE in the following format

$$\widehat{u}_{tt} + 2\widehat{u}_t + (1 + \omega^2)\widehat{u} = 0.$$

This is a second-order ODE, with characteristic equation

$$r^2 + 2r + (1 + \omega^2) = 0.$$

We find the complex roots for the equation with the quadratic formula

$$r = \frac{-2 \pm \sqrt{4 - 4(1 + \omega^2)}}{2} = -1 \pm i\omega.$$

The solution for the ODE is then

$$\widehat{u} = e^{-t} \left(\widehat{C}_1(\omega) e^{-i\omega t} + \widehat{C}_2(\omega) e^{i\omega t} \right), \quad (10)$$

where $\widehat{C}_1(\omega)_1$ and $\widehat{C}_2(\omega)_2$ are constant with respect to t . We use the following Fourier transform (can be found in the table in Krezig)

$$\mathcal{F} \left(\frac{\sin(x)}{x} \right) = \begin{cases} \sqrt{\frac{\pi}{2}} & \text{if } -1 \leq \omega \leq 1, \\ 0 & \text{else,} \end{cases} \quad (11)$$

to transform the initial conditions,

$$\widehat{u}(\omega, 0) = -\widehat{u}_t(\omega, 0) = \begin{cases} \sqrt{\frac{\pi}{2}} & \text{if } -1 \leq \omega \leq 1, \\ 0 & \text{else.} \end{cases} \quad (12)$$

By calculating $\widehat{u}(\omega, 0)$ and $\widehat{u}_t(\omega, 0)$ from (10), we have the following equations to solve (with $\widehat{u}(\omega, 0)$ and $\widehat{u}_t(\omega, 0)$ as in (12)):

$$\begin{aligned} \widehat{u}(\omega, 0) &= \widehat{C}_1(\omega) + \widehat{C}_2(\omega) \\ \widehat{u}_t(\omega, 0) &= -\widehat{C}_1(\omega) - \widehat{C}_2(\omega) - i\omega\widehat{C}_1(\omega) + i\omega\widehat{C}_2(\omega). \end{aligned}$$

First, we see that (12) only have real terms, meaning that $\widehat{C}_1(\omega) = \widehat{C}_2(\omega)$. Further, we find the expressions for $\widehat{C}_1$ and $\widehat{C}_2$ to be

$$\widehat{C}_1(\omega) = \widehat{C}_2(\omega) = \begin{cases} \frac{1}{2}\sqrt{\frac{\pi}{2}} & \text{if } -1 \leq \omega \leq 1, \\ 0 & \text{else.} \end{cases}$$

Next, we do an inverse Fourier transformation,

$$\begin{aligned} u(x, t) &= \int_{-\infty}^{\infty} e^{-t} \left(\widehat{C}_1(\omega)e^{-i\omega t} + \widehat{C}_1(\omega)e^{i\omega t} \right) e^{i\omega x} d\omega \\ &= e^{-t} \int_{-\infty}^{\infty} \left(\widehat{C}_1(\omega)e^{i\omega(x-t)} + \widehat{C}_1(\omega)e^{i\omega(x+t)} \right) d\omega \\ &= e^{-t} (C_1(x-t) + C_1(x+t)). \end{aligned}$$

$C_1(x)$ is found by inverse Fourier transform, and we again use (11) to find that

$$C_1(x) = \frac{\sin(x)}{2x}.$$

The function $u(x, t)$ is then

$$u(x, t) = \frac{e^{-t}}{2} \left(\frac{\sin(x-t)}{x-t} + \frac{\sin(x+t)}{x+t} \right).$$